

Fill prompt

Cancel

✓ Confirm

Placeholder 1

✕ Clear

Placeholder value

Placeholder 2

✕ Clear

Placeholder value

Placeholder 3

✕ Clear

Placeholder value

Preview

✕

The documentation presents in detail the Data Encryption Standard (DES), a symmetric encryption algorithm that was the NIST standard from 1977 to 2006, replaced by AES. It describes DES as a block algorithm that processes 64-bit plaintext blocks into 64-bit encrypted blocks. Attention is drawn to the different types of keys used in the algorithm: the main key (64-bit), the subkey (56-bit), and the 48-bit round keys generated for each of the 16 encryption rounds. The algorithm's operating mechanism is also explained, including initial and final permutation of plaintext, 16 encoding rounds, and operations within each round, such as block expansion, XOR operations with round keys, and the use of S-boxes for bit transformation. Additionally, the key processing procedure is described, including the creation of subkeys and round keys through permutations and cyclic bit shifts. The document concludes with a brief explanation of a working mode variant of the DES algorithm, the Electronic Codebook (ECB), characterized by constant mapping of plaintext blocks to ciphertext blocks for a given key.